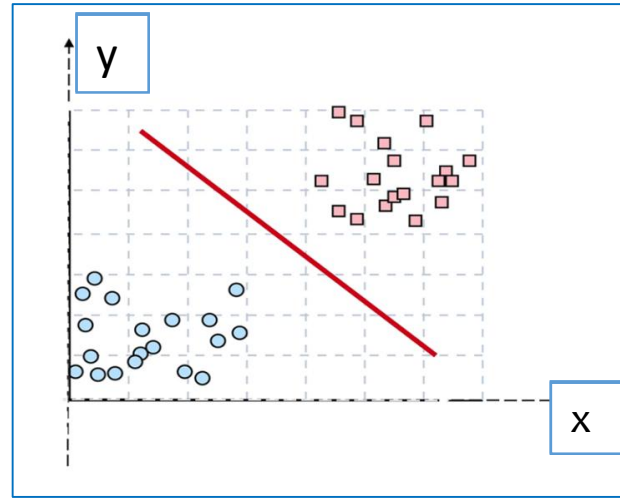


Equation of Straight Line

$$ax + by + c = 0$$



$$2x + 3y - 6 = 0$$

$$4x - y + 8 = 0$$

$$3x + 2y + 7 = 0$$

$$5x - 2y + 1 = 0$$

$$-1x + 2y - 2 = 1$$

Equation of Straight Line

$$ax + by + c = 0$$

$$ax_1 + bx_2 + c = 0$$

If $a = w_1$

$$b = w_2$$

$$w_1x_1 + w_2x_2 + c = 0$$

$$w_1x_1 + w_2x_2 = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \begin{bmatrix} x_1, x_2, \end{bmatrix}$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \begin{bmatrix} x_1, x_2, \end{bmatrix} = w^T x$$

$$w^T x + c = 0$$

Equation of Straight Line 3 Dimension

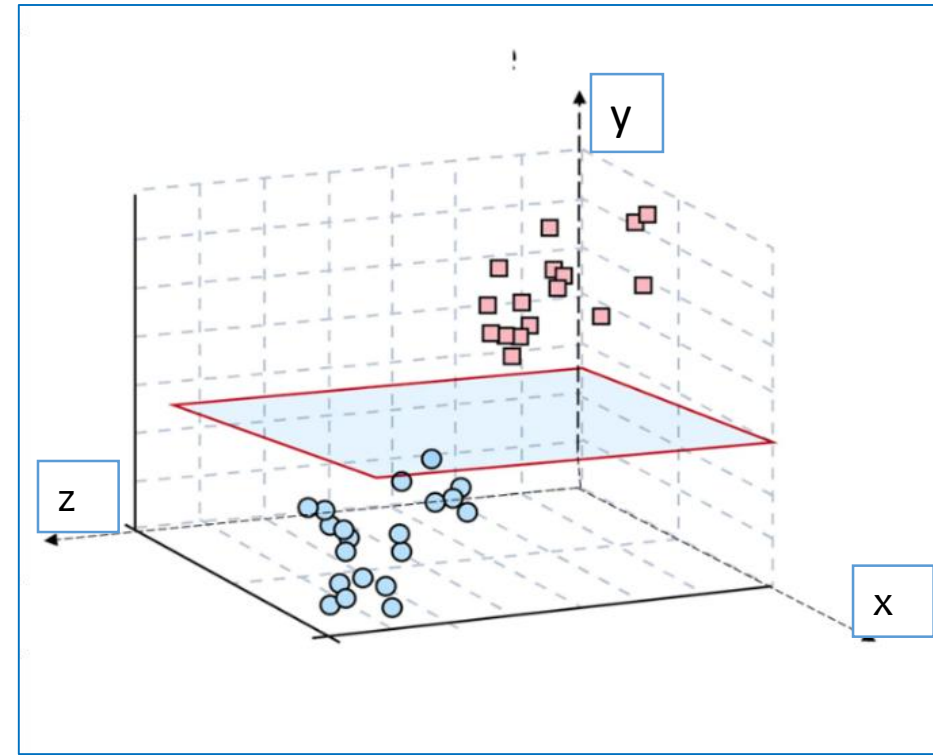
$$ax + by + cz + c = 0$$

$$ax_1 + bx_2 + cx_3 + c = 0$$

$$2x + 3y + 4z + 5 = 0$$

$$X - 2y + 3z - 6 = 0$$

$$x - 2y + 3z - 6 = 0$$



Equation of Straight Line

$$ax + by + cz + c = 0$$

$$ax_1 + bx_2 + cx_3 + c = 0$$

If $a = w_1$

$$b = w_2$$

$$c = w_3$$

$$w_1x_1 + w_2x_2 + w_3x_3 + c = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \begin{bmatrix} x_1, x_2, x_3 \end{bmatrix}$$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \begin{bmatrix} x_1, x_2, x_3 \end{bmatrix} = \mathbf{w}^T \mathbf{x}$$

$$\mathbf{w}^T \mathbf{x} + c = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 + c = 0$$

4D and Beyond

$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 \dots + w_nx_n + c = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 \dots \dots + w_nx_n = \begin{matrix} w_1 \\ \left[\begin{matrix} w_2 \\ w_3 \\ w_4 \\ \vdots \end{matrix} \right] \\ w_n \end{matrix} \left[\begin{matrix} x_1, x_2, x_3, x_4 \dots x_n \end{matrix} \right]$$

$$w^T x + c = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 \dots + w_nx_n + c = 0$$

Can I denote the same as below as well

$$w_1v_1 + w_2v_2 + w_3v_3 + w_4v_4 \dots + w_nv_n + c = 0$$

$$w^Tv + c = 0$$

$$y = m x + c$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y = 1 + 2$$

$$-1x + 2y = +3$$

$$-1x + 2y = 3$$

$$2y = 1x + 3$$

$$2y = 1x + 3$$

$$y = \frac{1}{2} x + \frac{3}{2}$$

$$y = \frac{1}{2} x + \frac{3}{2}$$

$$y = 0.5 x + 1.5$$

$$y = 0.5 x + 1.5$$

Equation of Straight Line

$$ax + by + c = 0$$

$$by + c = -ax$$

$$by = -ax + (-c)$$

$$y = \frac{-a}{b}x + \frac{-c}{b}$$

Equation of Straight Line

$$y = m x + c$$

m OR $\frac{-a}{b}$ = (Slope or Coefficient) slope of the line

c OR $\frac{-c}{b}$ = Intercept (where the best-fit line meets the Y axis).